Journal of Business Research 143 (2022) 1–15

Contents lists available at ScienceDirect

Journal of Business Research

journal homepage: www.elsevier.com/locate/jbusres

The effects of financial inclusion and the business environment in spurring the creation of early-stage firms and supporting established firms

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ARTICLE INFO

Keywords:
Early-stage firms
Established firms
Financial inclusion
Institutional environment
Panel VAR model

ABSTRACT

This study investigates the role of financial inclusion and the institutional environment in promoting entrepreneurship. Using the panel vector autoregressive model for a sample of 43 countries over the period from 2001 to 2018, we find that both financial inclusion and the institutional environment positively affect early-stage and established firms. In particular, our results reveal that basic financial products are important to stimulate the creation of early-stage firms, whereas access to credit supports established firms' viability. Our findings also indicate that lower business regulatory, tax and bureaucracy burdens; better access to commercial infrastructure; ease of entry to markets and government programs are essential for developing entrepreneurship. In particular, one of the most important implications of our findings is the vital role that government programs play in supporting entrepreneurship. Accordingly, several policies targeting promotion of entrepreneurship have been proposed and discussed.

1. Introduction

Recently, the interest of academic researchers, policymakers and governments in entrepreneurship has grown rapidly, mainly because of the positive expected effects in terms of accelerating innovation, enhancing productivity, creating new jobs and stimulating overall economic activity (Acs, Estrin, Mickiewicz, & Szerb, 2018; Audretsh & Keilbach, 2004; Pauwels, Clarysse, Wright, & Van Hove, 2016). Aware of the growing importance of entrepreneurship, governments, businesses and international organizations have devoted significant efforts to encourage the creation of new start-ups and to support established firms to persist in the market (Audretsh, Colombelli, Grilli, Minola, & Rasmussen, 2020). However, the results of such policies have not always been particularly successful. Early-stage and established firms are still facing serious difficulties in prospering because of two main constraints: (1) a lack of financial resources and (2) a weak business environment (Ali, Kelley, & Levie, 2020; Simón-Moya, Revuelto-Taboada, & Guerrero, 2014; Demirguc-Kunt, Klapper, Singer, Ansar, & Hess, 2018).

According to several studies, limited access to external finance is one of the main obstacles to the development of new and small businesses (Ayyagari, Demirguc-Kunt, & Makismovic, 2012; Beck, Demirguc-Kunt, & Martinez Peria, 2011). The constraint of access to finance is particularly noticeable in the case of start-ups and small firms because of a lack of collateralized assets and account histories (Wellalage, Hunja, Manita, Locke, 2021). These structural financial barriers make firms operate in an inefficient environment, where transaction costs, information asymmetries and uncertainty are relatively high (Fossen & Sorgner, 2021).

Considered one of the main pillars of the '2030 Sustainable Development Goals', financial inclusion, which is defined as the ease of access to financial services such as bank accounts, deposit accounts, credit and payment services for all participants in an economy (World Bank, 2014), is seen as an important factor that can spur entrepreneurial activities through several channels. First, financial inclusion provides accessible and affordable financial services to millions of 'unbanked' people (Barajas, Beck, Belhaj, & Ben Naceur, 2020), which encourages a wide range of potential entrepreneurs to start a new business and help them overcome their entry costs (Ayyagari, Demirguc-Kunt, & Makismovic, 2012). Second, financial inclusion enables established firms to have easy access to funds and credit, allowing them to survive and benefit from expansion opportunities (Banerjee, Breza, Duflo, & Kinnan, 2017). Third, with the emergence of digital financial services such as mobile money and financial technology applications (fintech), financial inclusion has facilitated the emergence of a new generation of innovative start-ups and the development of innovations in established firms (Elia, Margherita, & Passiante, 2020).

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https://doi.org/10.1016/j.jbusres.2022.01.014
Received 16 April 2020; Received in revised form 3 January 2022; Accepted 6 January 2022
Available online 25 January 2022
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However, until now, there has been no consensus in the literature on the impacts of financial inclusion on entrepreneurship (Leon, 2019). Although some studies have found strong evidence for a positive relationship (Allen, Demirguc-Kunt, Klapper, & Peria, 2012), other leading studies have shown that the impact of financial inclusion on the creation of new ventures is negligible and that all the benefits are oriented towards established firms (Crepon, Devoto, Duflo, & Pariente, 2011).

The second obstacle that early-stage and established firms face is an unfavorable local business environment (Aidis, Estrin, & Mickiewicz, 2012; Alvarez, Urbano, Coduras, & Ruiz-Navarro, 2011; Klapper, Laeven, & Raghuram, 2006). According to recent studies, a poor legal framework associated with heavy business regulation, high barriers to entry or inefficiently supportive public policies often affect the decision to start a project and discourage business opportunities (Bosma, Content, Sanders, & Stam, 2018; Djankov, La Porta, López-de-Silanes, & Shleifer, 2002; Dvouletý, Longo, Blažková, Lukeš, & Michal, 2018). The institutional literature has largely stressed that institutions can be a decisive factor in understanding the creation of new businesses and their growth and success (Ali, Kelley, & Levie, 2020). However, as many researchers have stated, the institutional environment can enhance or hinder the creation of new ventures and their sustainability (Baumol, 1990; Bosma, 2012).

The aim of this study was to address the above two gaps in the entrepreneurship literature by first examining whether financial inclusion fosters the creation of new ventures and/or sustains the viability of established firms. Second, we sought to provide a better understanding of the role that institutional conditions can play in enhancing the formation of new firms and supporting their persistence in the market.

For this purpose, we use the methodology of the Global Entrepreneurship Monitor (GEM), which assesses the process of creating new ventures and the survival of existing firms via two indicators: (1) total early-stage entrepreneurial activity (TEA) and (2) the established business ownership rate (Reynolds et al, 2005).

The GEM conceptual framework defines these two measures as the proportion of the adult population involved in creating a nascent firm, starting a new firm or running an operating firm (Levie & Autio, 2008; Reynolds et al., 2005). In this respect, following the GEM, the process of firm creation is measured by the TEA, and the persistence of established firms is measured by the established business ownership rate.

This study makes some important contributions to the entrepreneurship literature. First, although previous empirical findings on the relationship between financial inclusion and entrepreneurship are ambiguous and find mixed results, the present research seeks to provide a clear response to these divergent views. Similarly, whereas studies of the institutional environment's impact on entrepreneurial activity are scarce, mainly because of the complexity of the interactions among institutional factors (Ali, Kelley, & Levie, 2020; Simón-Moya, Revuelto-Taboada, & Guerrero, 2014), this study adds to the literature by giving a better understanding of the conditions that favor or damage the process of business creation and established firms' persistence. Second, unlike the extant studies that have focused mainly on the creation of new business activities, our research makes a considerable contribution to the literature by extending the investigation of the impacts of financial inclusion and the institutional environment on the survival of established firms. This later contribution is mainly motivated by the belief that (1) established firms are very important contributors to economic growth because they provide stable employment and long-term value for customers and stakeholders (GEM, 2020), and (2) financial inclusion and institutional conditions affect each stage of firm development (i.e., from birth to maturity) differently, which is critical when designing policies. Finally, compared with previous studies, which have mostly applied static approaches, we used the estimated impulse response functions and the forecast error variance decomposition under the panel vector autoregressive (PVAR) framework to examine the dynamic impact of financial inclusion and the institutional environment on the level of entrepreneurship. To the best of the author's knowledge, this study is the first to investigate such dynamic relationships.

The remainder of the study is organized as follows. Section 2 presents the theoretical channels and empirical evidence regarding the relationships of both financial inclusion and the institutional environment with the level of entrepreneurship. Section 3 presents the dataset and introduces the PVAR model. Section 4 interprets and discusses the results. Finally, Section 5 concludes the study and proposes some policies.

2. Literature review and hypothesis setting

According to the GEM model, entrepreneurship is a process comprising different phases based on a firm's life cycle, e.g., intending to create, starting and running a firm (Reynolds, et al, 2005; Levie & Autio, 2008). In this regard, the GEM model distinguishes between two main phases: the process of creating early-stage firms and the persistence of established firms. These two processes involve individuals[1] or entrepreneurs who are actively engaged in starting and/or running a firm (Bosma, 2012). Accordingly, the GEM model describes early-stage firms as new firms in which two types of entrepreneurs are engaged: (1) nascent entrepreneurs, who intend to start a firm, and (2) new entrepreneurs who own a firm that has operated for less than 3.5 yr. However, established firms are represented by entrepreneurs who have run a firm that has operated for more than 3.5 years and have survived the 'liability of newness' (Reynolds et al., 2005).

However, despite the growing importance of entrepreneurship in terms of job creation and economic growth (Acs et al., 2018; Bosma et al., 2018), only a few studies have examined the effects of financial inclusion on the creation of new firms and the viability of established firms (Leon, 2019). Moreover, the number of empirical studies examining the relationship between institutions and entrepreneurship are still limited in the literature, mainly because of the difficulties related to measuring institutional variables (Ali, Kelley, & Levie, 2020; Simón-Moya, Revuelto-Taboada, & Guerrero, 2014). Consequently, the following subsections will describe how financial inclusion and institutional environment may affect the entrepreneurship process.

2.1. Financial inclusion and entrepreneurship

2.1.1. Financial inclusion and the creation of early-stage firms

Theoretically, access to financial services stimulates the creation of new firms by mobilizing and allocating the necessary resources to entrepreneurs, helping them to take risks and start a business (King & Levine, 1993).

In particular, the availability of basic financial products such as bank accounts, deposits, savings or other payment services can reduce transaction costs, facilitate the transfer of wages and remittances, increase savings and open access to credit (Allen et al., 2012; Beck, Demirguc-Kunt, & Martinez Peria, 2011). In addition, shifting from cash payments into formal accounts solves the problem of information asymmetry, reduces uncertainty, increases the speed of transactions, improves efficiency and safety levels and encourages potential entrepreneurs to set up a business (Ayyagari, Demirguc-Kunt, & Makismovic, 2012; Levie & Autio, 2008). In this regard, Ashraf, Karlan and Yin (2009) showed that saving products tend to improve household decision-making and reinforce women's economic empowerment in business. Moreover, increasing the availability of bank branches encourages the development of small businesses, especially in rural and remote regions (Jiang et al., 2019). Finally, the literature has largely stressed that when starting a business, new entrepreneurs are often discriminated against by financial institutions because of lack of collateralized assets and account history. These financial constraints

[1] Note, that following the GEM concept, an entrepreneur is defined as an individual (an adult between 18 and 64 years old) who is actively engaged in setting up and/or running a firm (Reynolds et al., 2005; Levie & Autio, 2008).

limit the entrepreneurs' access to credit and inhibit their entry to the markets (Ayyagari, Demirguc-Kunt, & Makismovic, 2012; Leon, 2019; Wellalage et al., 2021). In this case, however, Allen et al. (2012) and Barajas et al. (2020) stated that whenever individuals face obstacles in access to credit, basic financial services often become an effective source of funding for new businesses and provide better results for enhancing ventures than access to credit.

Therefore, we can argue that early-stage firms are more likely to benefit from the availability of basic financial products than from access to credit. Thus, we can hypothesize that:

H1: Basic financial products are positively and significantly related to the emergence of early-stage firms.

2.1.2. Financial inclusion and established firms' viability

In contrast to the first hypothesis, some studies have raised doubts about the unconditionally positive effect of financial inclusion on the creation of new businesses and have shown that financial inclusion benefits established firms (Banerjee et al., 2017; Crepon et al., 2011).

According to Beck, Demirguc-Kunt and Martinez Peria (2011), and Fossen and Sorgner (2021), access to banking services encourages established firms to innovate at a faster rate, helping them to pursue growth opportunities and strengthen their activities. Recently, by using randomized evaluations, Banerjee et al. (2017) found that better access to credit enhances the activities of established firms, whereas early-stage firms remain excluded from the benefits of the availability of bank loans. Other empirical studies have found similar results and concluded that bank credit has a negligible impact on new business creation (Crepon et al., 2011, Karlan & Zinman, 2010), although existing firms benefit from most of the loans created by better access to financial services.

As a result, we can state that access to credit is expected to affect the viability of established firms positively and significantly. According to the analysis above, the following hypothesis is proposed:

H2: Access to credit is positively and significantly related to established firms' viability.

2.2. Institutional environment and entrepreneurship

In addition to the crucial role played by access to finance in promoting entrepreneurship, several other institutional factors have been identified in the literature as key factors for fostering the creation of new firms and increasing the survival of existing ones (Bosma et al., 2018; Noelia & Rosalia, 2020). Drawing on institutional theories, the GEM framework proposes nine institutional factors, known as entrepreneurial framework conditions (EFCs), which are expected to facilitate the process of entrepreneurship (Bosma et al., 2018). In this study, we focus our analysis on the four most important EFC variables[2]: (1) government regulation, taxes and bureaucracy; (2) access to commercial and professional infrastructure; (3) entry regulation and (4) government programs.

Government regulation, taxes and bureaucracy

Government regulation, taxes and bureaucracy are defined as a set of rules, laws and regulations that are implemented by the government to provide an appropriate environment for the creation of new ventures and the persistence of established firms (Bosma et al., 2012; Castaño, Méndez, & Galindo, 2015). If the environment is not appropriate, entrepreneurs will face more restrictions when starting a project and may miss several business opportunities (Baumol, 1990).

In this context, empirical studies have shown that excessive business regulation constitutes a barrier to promoting entrepreneurship (Ali, Kelley, & Levie, 2020; Alvarez, Urbano, Coduras, & Ruiz-Navarro, 2011). For instance, Van Stel, Storey and Thurik (2007) have indicated that the large number of administrative procedures undertaken to start a business and/or obtain a permit and license, and the long time they require, will have a negative impact on the formation of new ventures and business activity. Similarly, Djankov et al. (2002) found that the cost of bureaucratic delay, corruption and complicated formalities in opening a business hamper the creation of new firms. Another important obstacle caused by rigid government regulation is high taxes. According to Aidis et al. (2012), the tax burden reduces the financial returns and capital of firms, making them unable to innovate and prosper. Government actions may also affect the access to capital for entrepreneurs by limiting the independence of banks to ease access to credit and facilitate the free flow of funds (Levie & Autio, 2008).

As a result, previous studies have emphasized the importance of government regulation in providing an enabling environment, free from excessive business regulation, high taxes and bureaucracy burdens (Alvarez et al., 2011; Ali, Kelley, & Levie, 2020; Simón-Moya, Revuelto-Taboada, & Guerrero, 2014). We believe that such a business environment will nurture the creation of early-stage firms and allow established firms to pursue their activities in the market. From these arguments, we therefore propose the following hypothesis:

H3: Lower business regulation, tax and bureaucracy burdens are positively and significantly related to the emergence of early-stage firms and established firms' viability.

Commercial and professional infrastructure

The commercial and professional infrastructure reflects the availability and the quality of commercial and professional business services such as subcontractors, accountants, lawyers, bankers, business consultants and advertising. Access to these business services facilitates the operation of start-up firms and overcomes the gaps in their resources or competencies (Levie & Autio, 2008). Particularly, professional services tend to reduce transaction costs and increase efficiency in management, helping new and established firms to improve their competitiveness (Bosma, 2012).

We believe that both new and established firms will be significantly affected by changes in commercial and professional services. Accordingly, we can advance the following hypothesis:

H4: Easing access to commercial and professional services is positively and significantly related to the emergence of early-stage firms and established firms' viability.

Entry regulation

In addition to previous environmental conditions, entry regulation is another consistent factor that plays an important role in the process of creating new firms. Entry regulation, defined as the ease of entry of new firms into the markets, is related to the costs of the administrative requirements of starting a new business (Acs et al, 2018). Given the relevance of this environmental factor, many researchers have examined the effects of the barriers to entry on entrepreneurship (Aidis et al., 2012; Klapper, Laeven, & Raghuram 2006). According to Djankov et al. (2002), high entry regulation, such as excessive administrative requirements for starting a business and/or register a property, have an adverse effect on the entry of new small firms. Similarly, Klapper, Laeven and Raghuram (2006) described entry regulation as a 'form of protection', pushing large firms to rely on the subventions of public policy and pushing out new potential start-ups. In the same vein, Farinha, Lopes, Bagchi-Sen, Sebasti & Oliveira (2020) have found that market openness increases business competition and consequently allows entrepreneurs to have more opportunities to create their firms. For these reasons, we believe that easing the constraints to entry into the markets will encourage the creation of new firms. Therefore, we propose the following hypothesis:

[2] As highlighted by many researchers, the existence of multiple interrelations among the different institutional environment factors makes it empirically impossible to include all nine EFC variables simultaneously in one unique model (Castaño, Méndez, & Galindo, 2015; Simón-Moya, Revuelto-Taboada, & Guerrero, 2014; Ali, Kelley, & Levie, 2020). To overcome this problem, some scholars have recommended focusing on only a specific set of these conditions in order to help policymakers design better appropriate policies (Noelia & Rosalia, 2020). We have been also constrained by the PVAR model, where a maximum of nine endogenous variables can be included in the model.

H5: Easing the regulation of entry into markets is positively and significantly related to the emergence of early-stage firms.

Government programs

The fourth EFC considered in this study is government programs. Following the institutional theory, governments programs are defined as the different types of support for entrepreneurs such as business incubators, accelerators, innovation hubs, science parks and other initiatives that aim to promote new and innovative ventures (GEM, 2020; Pauwels, Clarysse, Wright, & Van Hove, 2016). These government initiatives provide a wide range of support to entrepreneurs, from office space, business support services, mentoring and networking services to direct financial subsidies, credit facilities and government equity crowdfunding (Mian, Lamine, & Fayolle, 2016).

As such programs are considered to be drivers of innovation, most countries today have implemented government programs and devoted large amounts of public resources to help the creation of new innovative firms and increase their chance of surviving (Audretsh et al., 2020; Noelia & Rosalia, 2020). Given the important investment of public resources dedicated to support entrepreneurship, sharp debates have taken place in the literature about the role of government programs in enhancing entrepreneurship. Whereas some researchers state that incubation programs contribute to enhancing the creation of new ventures (Audretch et al., 2020; Mian, Lamine, & Fayolle, 2016), others believe that these public policies are costly and not always effective in supporting new and innovative firms (Dvouletý et al., 2018; Luckes, Longo & Zouhar (2019). In this respect, Dvouletý et al. (2018) conducted research on Czech incubators and found that government programs were not effective in promoting entrepreneurship, since most new firms failed after receiving substantial support from the government. Luckes, Longo and Zouhar (2019) confirmed these results and concluded that governmental public policy programs have not been successful in helping new innovative start-ups.

Although government programs essentially target new start-ups, Audretsh et al. (2020) have shown that new ventures may generate significant positive spillovers into the local economy and on a national level. Thus, we believe that incubation programs will benefit both early-stage firms and established firms.

Consequently, we propose the following hypothesis:

H6: Supportive governmental programs are positively and significantly related to the emergence of early-stage firms and established firms' viability.

3. Methodology and data

3.1. Variables of interest

Following the GEM approach, entrepreneurship is defined as any attempt at new business or new venture creation, such as self-employment, a new business organization or the expansion of an existing business, by an individual, a team of individuals or an established business. As noted in Section 2.1, according to the GEM model, the entrepreneurship process comprises two phases: the creation of new ventures and established firms' viability. Therefore, in this study, in our definition of entrepreneurship, we refer to the process of creating new ventures and the persistence of established firms which involve individuals (entrepreneurs) who decide to start and/or manage a firm (Bosma, 2012).

Consequently, in our study, two proxies measure entrepreneurship: (1) the TEA and (2) the established business ownership rate (Reynolds et al, 2005). It is important to note that the main difference between these two measures is the age of the firm. According to the GEM definition, the two measures of entrepreneurship used in this study are as follows:

- The TEA is measured by the proportion of the adult population (18–64 years old) currently engaged in the creation of nascent firms and those who have been running new firms for less than 3.5 yr. This measurement recognizes the creation of new ventures and represents the dynamics of new business activity (Levie & Autio, 2008).
- The established business ownership rate is measured by the proportion of the adult population (18–24 years old) who have been actively managing a firm for more than 3.5 yr. This rate reflects the persistence of established firms and captures the extent to which they survive and prove to be economically viable for a period exceeding 3.5 yr.

The data for these two measures are derived from the GEM's Adult Population Survey[3]. In order to make the data comparable across countries, the same survey was administered at the same time of the year to a representative national sample targeting at least 2,000 randomly selected adults. The objective is to collect information on the adult population aged between 18 and 64 years involved in the process of establishing a new business and those who own and manage an operating firm (GEM).

3.2. Other variables

In addition to the two variables of interest presented above, we considered a set of 11 other endogenous variables classified into three groups: (1) four variables used as a measure of financial inclusion, (2) four variables used as measures for the business environment and (3) three macroeconomic variables included to control for heterogeneity and country-specific effects.

3.2.1. Financial inclusion variables

In this study, we used four proxies of financial inclusion: (1) the number of ATMs per 100,000 adults, (2) the number of bank branches per 100,000 adults, (3) domestic credit to the private sector as a percentage of gross domestic product (GDP), and (4) bank deposits as a percentage of GDP. The data for all these variables were extracted from the World Bank's Global Financial Inclusion Database. A detailed description of the variables is provided in Table 1.

3.2.2. The institutional environment

The second group of variables comprises the country's institutional environment variables. Specifically, we used the following four EFC variables: (1) bureaucracy and taxes, (2) commercial and professional infrastructure, (3) market openness and (4) government programs (see Table 1).

The data on the institutional environmental variables were selected from the National Expert Survey (NES). As a part of the GEM approach, the NES collects the views of experts on different dimensions of institutional environmental factors in a specific country. The NES sample comprises a minimum of 36 respondents, with four experts drawn from each of the EFC categories. Out of this sample, a minimum of 25% must be entrepreneurs or business owners, and 50% must be professionals. The responses to the items use a five-point Likert scale, where each variable represents an assessment ranging from 1, meaning that the statement is completely false according to the expert, to 5, meaning that the statement is completely true. Following the GEM theoretical framework, the better the state of the EFCs, the greater the likelihood that entrepreneurs will be able to create and develop their businesses.

3.2.3. Control variables (macroeconomic variables)

A set of three macroeconomic variables was used in this study to control for the heterogeneity among countries. These factors include real

[3] Since 1999, the GEM has collected data about entrepreneurship on an annual basis from a random national pool of adults (between 18 and 64 years of age) in each country (Reynolds et al., 2005). Annually, a national academic team or research institution leads the data collection process to ensure the quality and 'harmonization' of the data (Ali et al., 2020).

Table 1
Descriptions and definitions of the variables.

Variables	Definition	Symbol	Source	Sign	Explanation
Entrepreneurship measures					
The total early-stage entrepreneurial activity (TEA)	The proportion of the adult population (18–64 years old) who are currently engaged in the creation of nascent firms and those who are running new firms for less than 3.5 years.	*EntES*	GEM	N/A	This variable measures the rate of creation of new ventures and represents the dynamics of new business activity.
The established business ownership rate	The proportion of the adult population (18–64 years old) who has been actively managing a firm for more than 3.5 years.	*EntEB*	GEM	N/A	This variable measures the established firms' persistence process and captures to what extent established firms' survive and prove to be economically viable for a period exceeding 3.5 years..
Financial inclusion variables					
Number of ATM	Number of ATMs per 100,000 adults	*ATM*	WDI	+	This variable indicates how many people or firms have accounts at financial institutions.
Number of bank branches	Number of bank branches per 100,000 adults	*Branch*	WDI	+	More new bank branches in regions previously with no banks will support new businesses.
Private credit	Domestic credit to the private sector provided by banks (% of GDP)	*PCMD*	WDI	+	This variable measures the extent to which a country's financial system may allocate credit to entrepreneurs.
Bank deposits	Bank deposits (% of GDP)	*BDepo*	WDI	+	This variable represents 'demand, time and savings deposits in deposit money banks as share of GDP'.
Institutionnel Environmental variables: Entrepreneurial Framework Conditions					
Bureaucracy and taxes	Government policies	*BurTax*	GEM	+	This proxy captures government policies (legal regulation, administrative procedures, tax burdens, etc.) that might discourage prospective entrepreneurs.
Governmental Programs	Public Programs	*GovPro*	GEM	+	This variable represents specific initiatives for entrepreneurs such as incubators, accelerators, concrete territorial actions that may support new innovative startups.
Commercial and professional infrastructure	Property rights, commercial, accounting and other legal services.	*ComPI*	GEM	+	This captures a good commercial and services infrastructure that may stimulate new and growing firms.
Internal market openness	This represents the extent to which new firms are free to enter existing markets.	*IntMOP*	GEM	+	This reflects the ease of entry into the market. Low barriers will facilitate the entry of new firms.
Macroeconomic Variables					
Economic growth	The first difference of the logarithm of GDP (constant $).	*RGDP*	IMF-WEO	+/-	A greater economic activity creates positive economic expectation and increases business opportunities. In contrast, following GEM theory, countries with less economic growth have, in turn, higher rates of entrepreneurship.
Inflation rate	The first difference of the logarithm of the consumer price index.	*Inf*	IMF-WEO	–	A high inflation rate can erode profits and increase the cost of doing business.
Unemployment	The rate of unemployment, calculated as the share of the labor force that is without work.	*Unem*	WDI	+	Following GEM literature, higher unemployment leads to more entrepreneurial activity.

Note: This table explains the variables definitions, symbols employed, source of data, the expected signs and the economic interpretations behind the expected type of relationship.

GDP, inflation and the unemployment rate. Following Wennekers and Thurik (1999), developing countries are characterized by a high rate of necessity entrepreneurship, defined as people who start their own business because they have no jobs or no other alternatives. In this regard, we expect that the economic growth variable will have a negative effect on the level of entrepreneurship. However, some authors state that greater economic activity also creates positive economic expectations and improves business opportunities (Castaño, Méndez, & Galindo, 2015). In this case, the economic growth variable could be positive.

Regarding the level of unemployment, we expect that a higher unemployment rate will lead to more entrepreneurial activity. Finally, we expect that a high inflation rate will erode profits and increase the costs of doing business.

The data for these macroeconomic indicators were collected from the World Development Indicators Database. Table 1 provides detailed definitions of these variables.

3.3. Panel VAR model

Unlike previous studies, this study used the PVAR model, which has several advantages over traditional econometric models. First, the PVAR model has the ability to assess the duration of the impacts of shocks via impulse response functions (IRFs). Second, it has the advantage of assessing the contribution of each variable to the variance decomposition of all the other variables via forecast error variance decomposition (FEVD). Third, the use of the PVAR model was motivated by its ability to investigate the direction of the causality between the variables in the model. Finally, in addition to the panel and multivariate dimensions that this model includes, the PVAR model has the ability to account for unobserved individual heterogeneity in a multivariate context. A further benefit of the PVAR model is the ability to allow for the existence of country and time fixed effects, which lead to better consistency in the results (Abrigo & Love, 2016).

However, despite these several advantages, only a few studies have used this model in the entrepreneurship literature, as almost all of them have focused on the relationship between entrepreneurship and unemployment from different perspectives (Cebula, Davis, Koch, & Saunoris, 2020).

The general regression form of the PVAR model has the following vector–matrix representation:

$$\boldsymbol{Y}_{it} = \boldsymbol{A}_0 + \boldsymbol{A}_1 \boldsymbol{Y}_{it-1} + \boldsymbol{A}_2 \boldsymbol{Y}_{it-2} + \cdots + \boldsymbol{A}_p \boldsymbol{Y}_{it-p} + \boldsymbol{u}_i + \boldsymbol{\lambda}_t + \boldsymbol{e}_{it}$$

where $i \in [1, 2, \cdots, N]$ and $t \in [1, 2, \cdots, T]$. Here, $Y_{it} = (Ent_{it}, FInc_{it}, \boldsymbol{X}_{it}, \boldsymbol{Z}_{it})$ is the vector of endogenous variables, which are all assumed to be stationary. *Ent* is the entrepreneurship variable (proxied by the prevalence of early-stage firms (*EntES*) or the prevalence of established firms (*EntEB*)). *FInc* refers to the financial inclusion variable, which is proxied by one of the four variables (*ATM*, *Branch*,*PCDM* and *BDepo*). $\boldsymbol{X}$ is a vector of institutional environment variables formed by the four variables *BurTax*, *ComPI*, *IntMOP* and *GovPro*. Finally, the vector $\boldsymbol{Z}$ is a vector

of macroeconomic variables that includes *RGDP*, *Unem* and *Inf*. The matrix A_j includes the parameters to be estimated, and u_i, λ_t and e_{it} designate the specific panel fixed effects, time fixed effects and idiosyncratic errors, respectively.

The pre-estimation, estimation and post-estimation procedures of the PVAR model can be summarized in seven steps as follows:

First, before estimating the PVAR model, we checked the stationarity of the different variables in our framework by using four panel unit root tests. These tests include two Augmented Dickey–Fuller Fisher tests (namely, Fisher's chi-square and Choi's Z-stat), and the Levin et al. (2002) and the Im et al. (2003) panel unit root tests. All these four-panel unit root tests assume cross-sectional independence and non-stationarity under the null hypothesis.

Second, we used three moment information criteria (the Bayesian information criterion (MBIC), the Akaike information criterion (MAIC) and the Hannan and Quinn information criterion (MQIC)) to select the optimal lag length for the PVAR model.

Third, we estimated the PVAR model using the generalized method of moments and by instrumenting the endogenous variables using lags from 1 to 4[4] with the optimal lag length as selected in Step 2.

Fourth, we checked for the stability of the estimated PVAR model by ensuring that all the moduli of the companion matrix of the estimated PVAR model were strictly less than one. This is an important condition, especially for the interpretation of impulse–response functions (IRFs) and forecast error decomposition.

Fifth, we used the IRFs to analyze the responses of the two entrepreneurship variables to a one standard deviation (1-SD) shock affecting the four financial inclusion proxies separately[5].

Sixth, we used the FEVD to analyze the contribution, as a percentage, of exogenous shocks to the variance of all the variables in the system for up to 10 periods (10 years).

Finally, we analyzed the causality direction between the prevalence of entrepreneurship and both the financial inclusion and institutional environmental variables.

4. Results and discussion

4.1. Preliminary analysis

4.1.1. Descriptive statistics and correlation analysis

Table 2 reports the descriptive statistics and the correlation analysis for the 13 variables in our system. Panel A of Table 2 shows that the average prevalence of entrepreneurship in our sample is 10.01% for early-stage firms and 6.63% for established firms. The maximum and minimum values for early-stage firms correspond to Peru and Japan in 2004, respectively. Concerning established firms, the maximum and minimum values were found for Uganda in 2013 and Ecuador in 2010, respectively.

Regarding the financial inclusion proxies, we found that the average number of ATMs per 100,000 adults was 75.31, the average number of bank branches per 100,000 adults is 25.09 banks and the average domestic credit to the private sector as a percentage of GDP is 85.47%. Finally, the average bank deposits as a percentage of GDP is 66.39%. In addition, we found that the maximum values for ATMs, bank branches, domestic credit to the private sector and bank deposits were registered for Korea in 2013, Spain in 2007, Iceland in 2006 and Japan in 2016, respectively. The minimum values were registered for Uganda in 2004 and 2014, Slovenia in 2004 and Uganda in 2001, respectively.

Regarding the business environment variables, the descriptive analysis shows that the average values of *BurTax*, *IntMOP*, *ComPI* and *GovPro* are not high on the 1–5 scale. The highest average value was found for the *IntMOP* variable (= 3.089), and the lowest average value (= 2.396) is found for the *BurTax* variable. The maximum values of *BurTax*, *IntMOP*, *ComPI* and *GovPro* were found for Iceland in 2006, Singapore in 2001, the U.S. in 2004 and Germany in 2009, respectively. The minimum values of *BurTax*, *ComPI* and *GovPro* were found for Argentina in 2002, 2004 and 2016, respectively, and Greece in 2003.

Finally, Panel B of Table 2 reports the results of the estimated correlation coefficients between the two entrepreneurship variables (*EntES* and *EntEB*) and the variables of the other three groups of variables (financial inclusion, business environment and macroeconomic variables). The results show that although *EntES* is positively and significantly correlated with all the variables in the model, *EntEB* is only positively correlated with a few variables (unemployment, domestic credit to the private sector, bank deposits, *BurTax*, *IntMOP* and *GovPro*).

4.1.2. Graphical analysis

For a better understanding of the relationship between financial inclusion and the prevalence rate of entrepreneurship, in Fig. 1, we plot the yearly mean averages of the prevalence of early-stage firms versus the yearly mean averages of the four financial inclusion proxies separately. The results show evidence for a robust positive association (a positive slope of the fitted linear regression line (red line)) between the prevalence of early-stage firms and three out of the four financial inclusion measures (number of ATMs, credit to the private sector and bank deposits). In particular, Fig. 1 shows evidence of an increasing association between the prevalence of entrepreneurship and financial inclusion over time, indicating an improvement in both *EntES* and the level of financial inclusion over time.

For the case of established firms, in Fig. 2, we report the association between the country averages for the prevalence of established firms and the country averages of the four financial inclusion variables. Overall, we found that the results depend on the financial inclusion variable considered. For instance, although the top left box of Fig. 2 shows the absence of a significant correlation between the prevalence of established firms and ATMs, a negative correlation was found for the number of bank branches (top right box of Fig. 2). Regarding the two other financial inclusion measures (domestic credit to the private sector and bank deposits), a strong positive correlation was found between these two entrepreneurship measures and the prevalence of established firms. In particular, our results show that Uganda and Thailand were the countries with the highest levels of established firms and lowest levels of financial inclusion.

4.2. Stationarity, order selection and stability

The results from applying the four panel unit root tests are reported in Table 3. We found that all the statistics have a *p*-value lower than the level of significance at 5%, except in a few cases. Specifically, 10 out of 13 series in our study were found to be stationary according to the four test statistics, and the other three series (*ATM*, *Branch* and *BDepo*) were found to be stationary according to two out of the four panel unit root tests. On the basis of these results, we confidently continued our analysis by estimating the PVAR model.

Table 4 shows that the optimal lag corresponds to a value of $k = 1$(see the MAIC, MBIC and MQIC). Therefore, first-order PVAR models were used in the remainder of our analysis.

Regarding the results on the stability of the estimated model (which can be obtained upon request), we found that the modulus of each

[4] We account for the country fixed effects by transforming each variable with the forward orthogonal deviations proposed by Arellano and Bover (1995), which was proven to be more robust than first-difference transformation.

[5] To ensure the orthogonalization of shocks when using Cholesky decomposition, we used the approach of Sims (1980), where the more exogenous variables appear earlier in the order, followed by variables that are more endogenous. The degree of exogeneity of the variables is mainly based on economic theory and previous studies in the field. As a result, the following order was used: institutional environment, financial inclusion variables, inflation, unemployment, and economic growth and entrepreneurship variables. The IRFs were obtained via 1000 Monte Carlo simulations.

Table 2
Descriptive statistics and correlation.

	EntES	*EntEB*	*RGDPG*	*Inf*	*Unem*	*ATM*	*Bra*	*DCPS*	*BDepo*	*BurTax*	*ComPI*	*IntMOP*	*GovPro*
Panel A: Descriptive statistics													
Mean	10.01	6.631	2.711	3.843	8.479	75.31	25.09	85.47	66.39	2.396	3.089	2.679	2.649
Std. Dev.	6.437	5.158	9.275	5.034	5.916	53.72	17.12	49.93	35.78	0.580	0.392	0.394	0.490
Min	1.480	0.330	- 15.55	- 1.676	0.489	1.099	1.121	0.186	11.30	1.220	1.940	1.820	1.340
Max	40.27	36.09	22.41	54.246	33.47	288.63	104.2	308.9	218.2	4.270	4.210	4.560	3.710-
Panel B: Correlation													
EntES	1	0.315***	0.247***	0.209***	0.215***	0.209***	0.361***	0.324***	0.413***	0.136***	0.233***	0.199***	0.270***
EntEB	–	1	0.058	0.036	0.238***	0.016	- 0.133**	0.132***	0.072**	0.099**	0.069	0.108**	0.132***

Note: This table reports the standard descriptive statistics including the mean, standard deviation (Std. Dev.), Minimum (Min), and Maximum (Max).

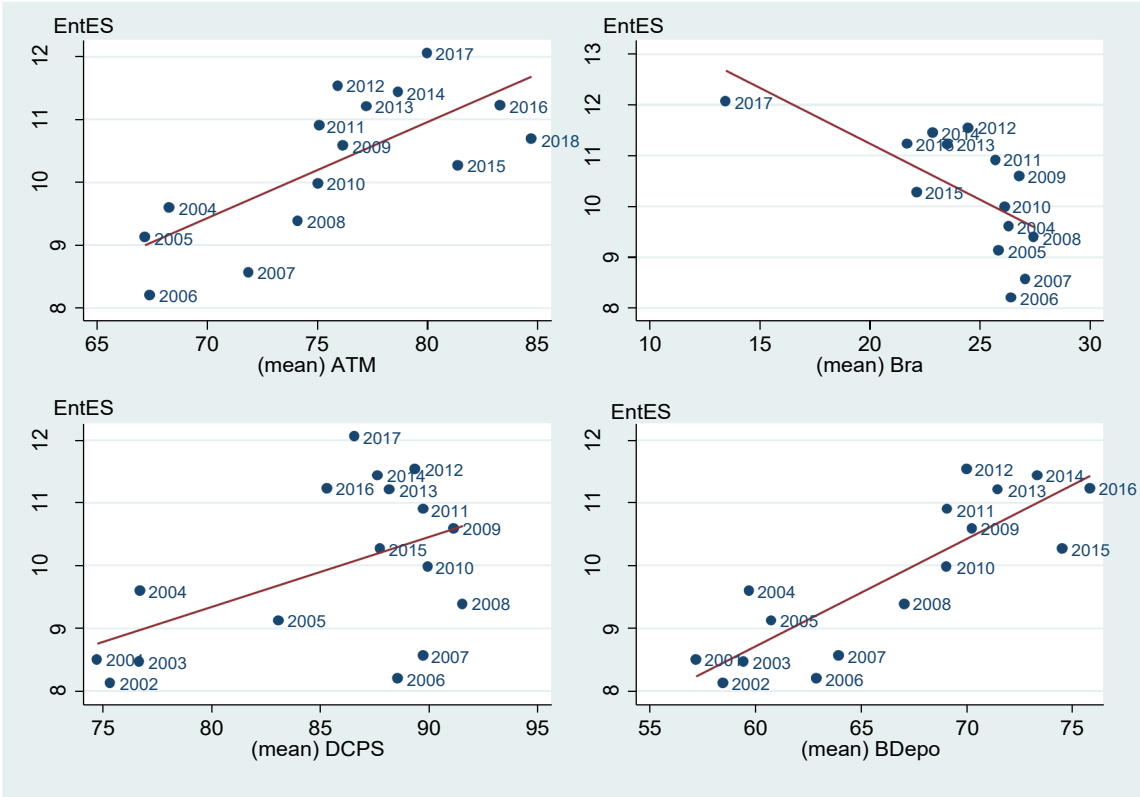

Fig. 1. Early stage firms and financial inclusion – Yearly means.

eigenvalue lies inside the unit root circle, which clearly indicates the stability of our estimated model (Kilian & Lütkepohl, 2017).

4.3. PVAR estimation results

4.3.1. Early-stage firms

As we can see in Table 5 below, our findings show evidence that financial inclusion fosters the creation of nascent and new firms (early-stage firms) for three out of the four financial inclusion proxies (*ATM*, *DCPS* and *BDepo*). In particular, we found that bank deposits had the highest positive effect ($\widehat{\beta} = 0.156^{***}$) on early-stage firms, which was more than double the effect of the number of ATMs ($\widehat{\beta} = 0.065^{***}$) and four times higher than that of domestic credit to the private sector ($\widehat{\beta} = 0.039^{***}$). The values of these coefficients indicate that, on average and holding all the other variables constant, an increase of 10 in the number of ATMs per 100,000 adults in year t will increase the rate of early-stage firms by 0.39% in year $t+1$. Similarly, a 10% increase in domestic credit to the private sector as a percentage of GDP at time t will increase the prevalence of early-stage firms at time $t+1$ by 0.65%. Finally, an increase in bank deposits as a percentage of GDP by 10% in year t will increase the prevalence rate of early-stage firms by 1.56% in year $t+1$. Overall, these results are consistent with the descriptive analysis in Section 4.1.2 (see Fig. 1), where a positive association was found for these three financial inclusion proxies.

Regarding the institutional environment variables, we found that 13 out of the 15 coefficients associated with the first lag of *BurTax*, *ComPI*, *IntMOP* and *GovPro* were significant (i.e., 10 coefficients were significant at 1%, two coefficients were significant at 5% and one coefficient was significant at 10%). Only the two coefficients associated with the first lag of *BurTax* in Models 1 and 3 were found to be insignificant. As expected, we found that the four institutional environment variables facilitate the creation of new business ventures, with the exceptions of *BurTax* in all the models, *ComPI* in Model 3 and *IntMOP* in Model 2. The negative sign of the coefficient associated with *BurTax* can be explained by the high rate of necessity entrepreneurship, where despite high entry barriers and complex administrative procedures, a large percentage of the population is forced to start a business because there are no other options for employment (Alvarez et al., 2011; Simón-Moya et al., 2014).

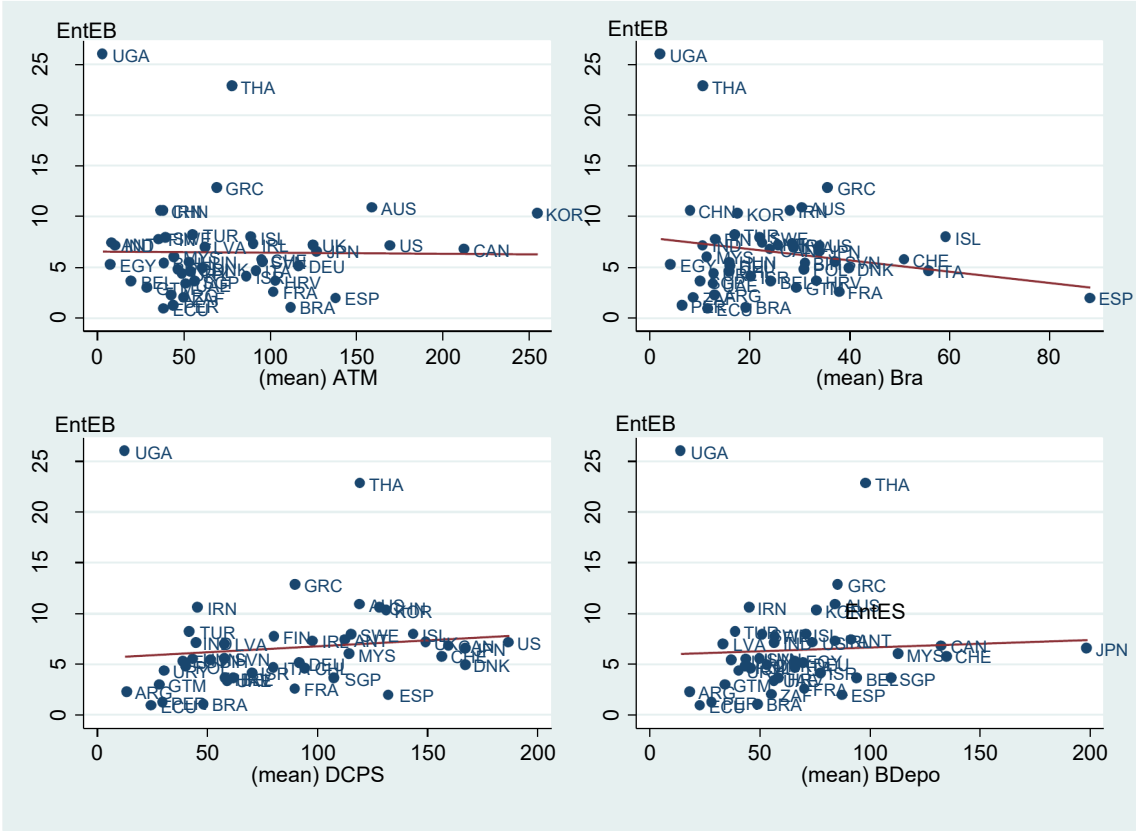

Fig. 2. Established firms and financial inclusion – Country means.

Table 3
Results of panel unit root tests.

	ADF-Fisher		LLC	IPS
	Chi-square	Z-stat		
EntES	167.420	- 4.169	- 4.903	- 4.433
	(0.000)	(0.000)	(0.000)	(0.000)
EntEB	178.420	- 5.550	- 21.272	- 6.807
	(0.000)	(0.000)	(0.000)	(0.000)
RGDPG	277.954	- 10.116	- 9.052	- 10.584
	(0.000)	(0.000)	(0.000)	(0.000)
Inf	245.179	- 8.808	- 12.626	- 7.507
	(0.000)	(0.000)	(0.000)	(0.000)
ATM	128.571	- 0.255	- 4.378	0.345
	(0.003)	(0.399)	(0.000)	(0.635)
Bra	127.284	1.046	- 3.012	0.466
	(0.004)	(0.852)	(0.000)	(0.679)
DCPS	122.993	- 1.936	- 6.780	- 2.384
	(0.005)	(0.026)	(0.000)	(0.009)
BDepo	117.493	- 2.036	- 1.357	1.739
	(0.014)	(0.009)	(0.087)	(0.959)
Burtax	164.983	- 4.875	- 9.228	- 4.967
	(0.000)	(0.000)	(0.000)	(0.000)
ComPI	230.976	- 7.507	- 7.751	- 12.056
	(0.000)	(0.000)	(0.000)	(0.000)
IntMOP	235.009	- 7.483	- 13.648	- 7.413
	(0.000)	(0.000)	(0.000)	(0.000)
GovPro	129.390	- 6.833	- 10.455	- 6.104
	(0.000)	(0.000)	(0.000)	(0.000)

Note: Table 2 provides the results of panel unit root tests.
(.) p-value is into parenthesis.

Interestingly, we found that only the government program variables (*GovPro*) had a positive expected sign and was highly significant in all four models. The results show and confirm the important role that the business environment plays in fostering the creation of new ventures.

4.3.2. *Established firms*

As we can see in Table 6, our results show a positive significant association between the two financial proxies (domestic credit to the private sector and bank deposits) and the prevalence of established firms. In particular, we found that the impact of those financial variables was approximately similar to that found in the case of early-stage firms. We found that following a 10% increase in domestic credit to the private sector and bank deposits, the prevalence of established firms will increase by 0.54% and by 1.29%, respectively. These results indicate that domestic credit to the private sector and bank deposits significantly support the viability of established firms and help them not fail. In contrast, we found that ATMs and the number of branches negatively affected the prevalence of established firms.

The results also show that the institutional environment variables (*BurTax*, *ComPI* and *GovPro*)[6] have a positive and significant effect on the prevalence of established firms in only 7 out of the 11 cases considered. In particular, we found that *BurTax* was positive and significant in both Models 2 and 4, highlighting the important roles of government policies, taxes and bureaucracy in maintaining the viability of established firms. The *ComPI* variable was found to have a positive and significant impact on the prevalence of established firms in only two models (Models 3 and 4). Finally, we found that government programs are an important factor in promoting and sustaining established firms, and were significant in three out of the four cases. These results confirm the importance of supportive government programs for both new venture creation and the survival of established firms.

Finally, the estimation shows that the impacts of economic growth, unemployment and inflation rates on early-stage and established firms were mixed in terms of their significance and signs. For early-stage

[6] We found that market openness is a determining factor only for early-stage firms.

Table 4
Results of selection order criteria.

EntES equations								
	Model 1		Model 2		Model 3		Model 4	
	$p = 1$	$p = 2$	$p = 1$	$p = 2$	$p = 1$	$p = 2$	$p = 1$	$p = 2$
MBIC	**−200.473**	- 92.516	**−214.335**	- 112.467	**−275.040**	- 149.166	**−231.201**	- 134.087
MAIC	**−57.382**	- 37.546	**−67.449**	- 47.329	**−80.284**	- 61.703	**−73.267**	- 54.851
MQIC	**−113.485**	- 56.339	**−125.662**	- 67.829	**−149.708**	- 100.025	**−135.841**	- 94.727
EntEB equations								
	Model 1		Model 2		Model 3		Model 4	
	$p = 1$	$p = 2$	$p = 1$	$p = 2$	$p = 1$	$p = 2$	$p = 1$	$p = 2$
MBIC	**−199.274**	- 91.881	**−188.215**	- 86.967	**−218.707**	- 112.473	**−209.018**	- 108.387
MAIC	**−58.054**	- 36.402	**−49.510**	–33.419	**−61.429**	- 42.306	**−56.525**	- 34.614
MQIC	**−115.051**	- 56.793	**−106.807**	- 53.681	**−113.026**	- 62.289	**−109.070**	- 58.417

Note: This Table reports the results of the three selection criterion namely MBIC, MAIC and MQIC for $p = 1$ and $p = 2$. The optimal order is the order for which these criterions are minimal.

Table 5
PVAR results estimation for the early stage entrepreneurship specifications.

Dependent variable	$EntES_t$			
Independent variables	Model 1	Model 2	Model 3	Model 4
$EntES_{t-1}$	0.266**	0.784***	0.634***	0.752***
	(0.107)	(0.081)	(0.044)	(0.059)
$RGDPG_{t-1}$	0.075*	0.099***	0.094***	0.122***
	(0.032)	(0.032)	(0.029)	(0.030)
$Unem_{t-1}$	0.246***	0.149***	–	0.201***
	(0.069)	(0.064)		(0.043)
Inf_{t-1}	–	0.104*	0.137***	0.375***
		(0.060)	(0.038)	(0.046)
ATM_{t-1}	0.065***	–	–	–
	(0.019)			
Bra_{t-1}	–	- 0.012	–	–
		(0.029)		
$DCPS_{t-1}$	–	–	0.039***	–
			(0.007)	
$BDepo_{t-1}$	–	–	–	0.156***
				(0022)
$BurTax_{t-1}$	- 3.131***	- 0.608	- 0.757	- 1.451***
	(1.152)	(0.760)	(0.511)	(0.542)
$ComPI_{t-1}$	3.479***	5.403**	- 3.011***	6.438***
	(1.207)	(01.088)	(0.478)	(0.659)
$IntMOP_{t-1}$	2.475*	- 3.667***	1.041**	–
	(1.352)	(1.308)	(0.420)	
$GovPro_{t-1}$	6.910***	3.828***	3.941***	2.248***
	(1.063)	(0.687)	(0.668)	(0.563)
No. of Panels	175	183	297	209
No. of Obs	34	34	38	33
GMM weight matrix	Robust	Robust	Robust	Robust

Note: We report only the equation interest, the entrepreneurship equation, of the PVAR model.
* Stands for 10% level of significance.
** Stands for 5% level of significance.
*** Stands for 1% level of significance. (.) Stands for standard errors.

firms, the impacts were positive and highly significant in almost all cases. Although a positive effect was only expected for the case of the economic growth and unemployment rate, the unexpected sign of the inflation rate was consistent with many previous studies (Leon, 2019). For established firms, we found that the coefficient associated with the inflation rate was significant in the four cases but had the expected negative sign in only two cases (number of bank branches and bank deposits). Similarly, we found that the coefficient associated with real economic growth was significant with an unexpected positive sign in three cases (ATM, domestic credit to the private sector and bank deposits).

Table 6
PVAR results estimation for the early stage entrepreneurship specifications.

Dependent variable	$EntEB_t$			
Independent variables	Model 1	Model 2	Model 3	Model 4
$EntEB_{t-1}$	0.239***	0.150***	0.188**	0.294***
	(0.071)	(0.048)	(0.093)	(0.064)
$RGDPG_{t-1}$	0.077**	- 0.016	0.257***	0.127***
	(0.035)	(0.018)	(0.059)	(0.048)
Inf_{t-1}	0.064*	- 0.100***	- 0.179***	0.212***
	(0.038)	(0.019)	(0.076)	(0.035)
ATM_{t-1}	- 0.023*	–	–	–
	(0.013)			
Bra_{t-1}	–	- 0.056***	–	–
		(0.022)		
$DCPS_{t-1}$	–	–	0.054***	–
			(0.013)	
$BDepo_{t-1}$	–	–	–	0.129***
				(0.025)
$BurTax_{t-1}$	- 1.132	1.545***	1.183	6.582***
	(0.808)	(0.499)	(0.937)	(0.777)
$ComPI_{t-1}$	–	0.266	4.743***	8.748***
		(0.781)	(1.433)	(0.864)
$GovPro_{t-1}$	6.382***	1.559***	5.762***	0.697
	(1.413)	(0.517)	(1.339)	(0.0984)
No. of Panels	168	155	152	179
No. of Obs	35	36	35	35
GMM weight matrix	Robust	Robust	Robust	Robust

Note: We report only the equation interest, the entrepreneurship equation, of the PVAR model. * Stands for 10% level of significance.
** Stands for 5% level of significance.
*** Stands for 1% level of significance. (.) Stands for standard errors.

4.4. Results for IRFs

Fig. 3 (Fig. 4) reports the results of the IRFs of *EntES*(*EntEB)* to a 1-SD shock to *ATM*, *Bra*, *DCPS* and *BDepo*. Fig. 3 shows that the response of *EntES* to a 1-SD innovation shock on the financial inclusion proxies was positive and significant for the cases of *DCPS* and *BDepo*. Regarding these two financial inclusion proxies, although the magnitude of the impact was approximately similar, we found that the duration of the impact of the shocks to *DCPS* lasts 2 yr longer than shocks to *BDepo* (6 yr versus 4 yr). For *ATM* and *Bra*, although the impact was positive during the first 5 to 6 periods, we found that it is insignificant since the zero-horizontal axis falls into the error band.

For *EntEB*, the IRFs reported in Fig. 4 show that the reactions of established firms to a 1-SD shock to the four financial inclusion variables are positive and statistically significant for the *DCPS* and *BDepo* cases but

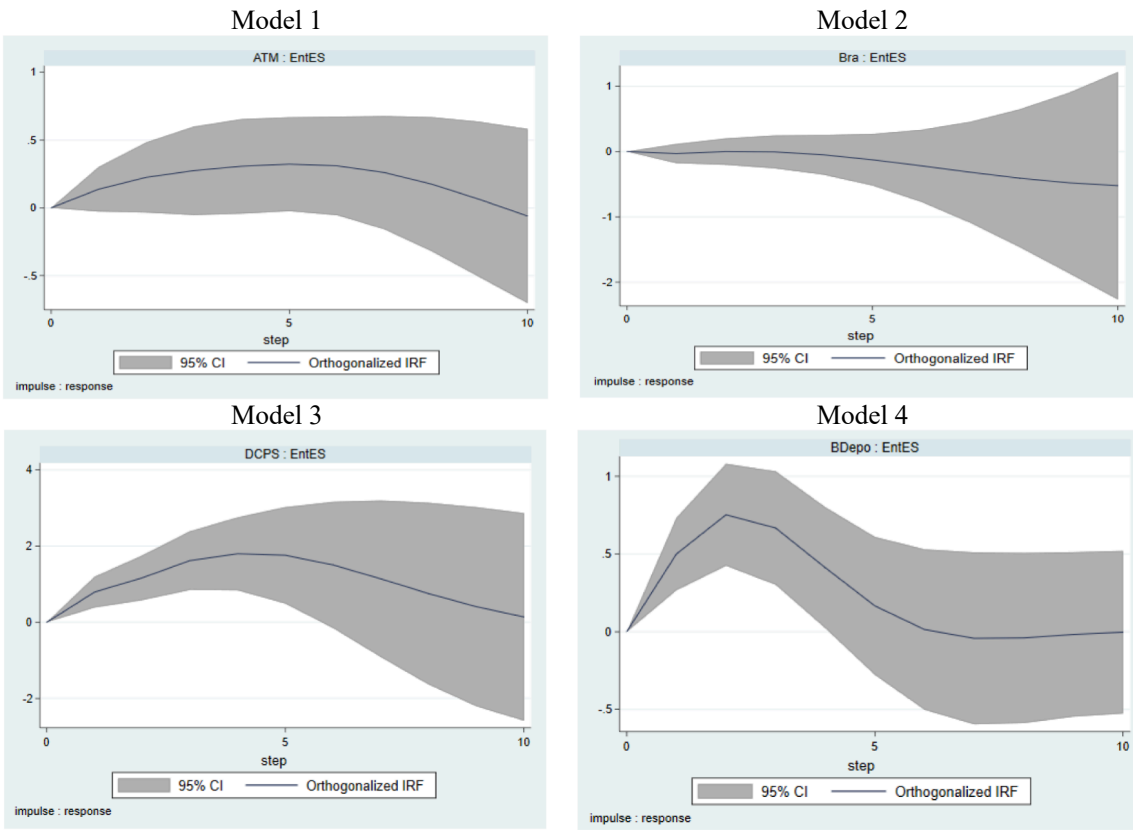

Fig. 3. IRFs of *EntES* to one standard innovation on *ATM, Bra, DCPS,BDepo.*

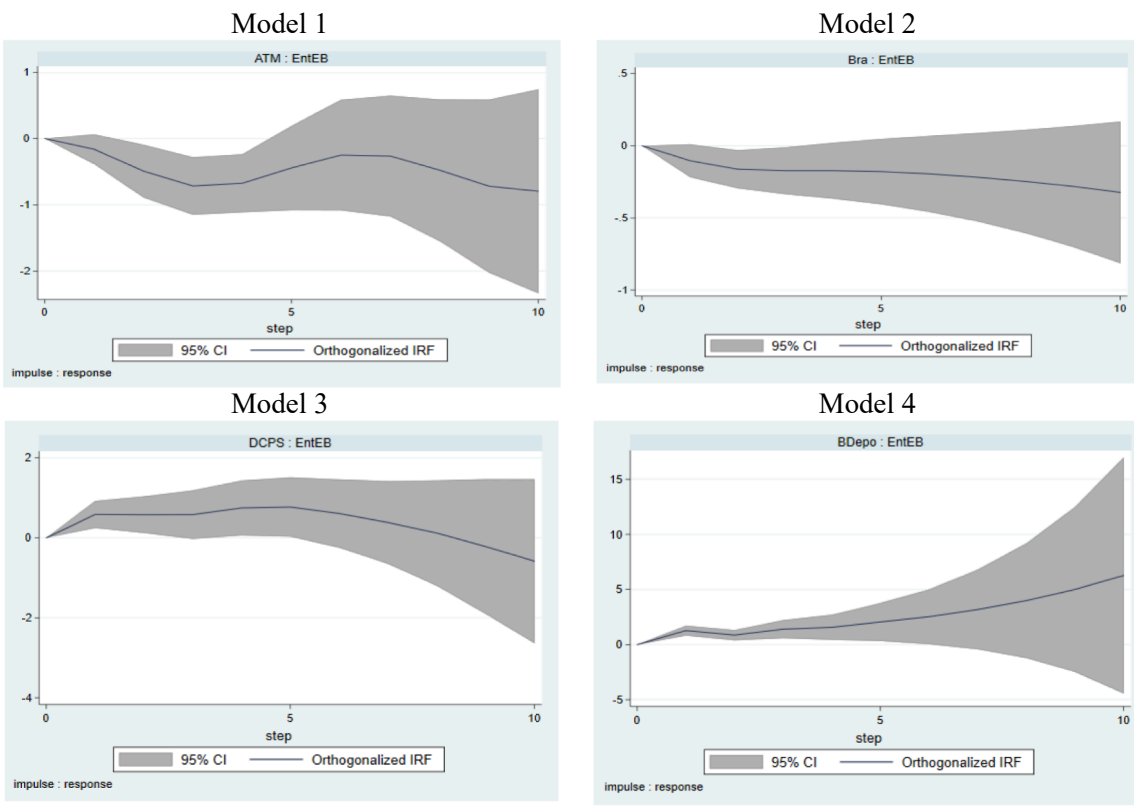

Fig. 4. IRFs of *EntEB* to one standard innovation on *ATM*, *Bra*, *DCPS,BDepo*.

significantly negative for *ATM* and *Bra*. The results show that following a shock to *ATM* and *Bra*, the reactions of established firms are very weak in magnitude and duration. For instance, a 1-SD shock to *ATM* and *Bra*, separately, lasts for less than one period (1 yr) for the case of *ATM* and is only significant during the second period (Year 2) for *Bra*.

4.5. Results for FEVD

Regarding the FEVD results (see Figs. 5 and 6), our findings show that the contribution of the financial inclusion variables to the variability in early-stage and established firms fluctuated significantly over time. In particular, we found that *DCPS* has the highest contribution (between 7% and 29% for *EntES* and between 18% and 42% for *EntEB*). In the second position, *BDepo* has a contribution that varies between 2% and 6% for *EntES* and between 6.5% and 9.4% for *EntEB*. In terms of the other two financial inclusion variables, their contribution to the variability in *EntES* and *EntEB* is weak. For instance, we found that the contribution of *ATM* to the variability of both *EntES* and *EntEB* was between 2% and 5%. For *Bra*, the contribution varies between approximately 0% and 13%.

However, the findings show that the joint contribution of financial inclusion and institutional environment variables to the variability of the prevalence of entrepreneurship is high overall: between 12% and 35% for *ATM*, between 4% and 51.5% for the number of bank branches, between 14% and 78% for domestic credit to the private sector, and between 14% and 55% for bank deposits.

4.6. Causality results

The results of the Granger causality tests are particularly helpful when designing policies to promote and stimulate the level of entrepreneurial activities in a country. A summary of the results is reported in Table 7 for *EntES* and in Table 8 for *EntEB*.

The results provide strong evidence for unidirectional causality from three financial inclusion proxies (*ATM*, *DCPS* and *BDepo*) to *EntES* at the 5% significance level. Evidence for bidirectional causality was found only for the two cases of *ATM* and *BDepo* at the 1% significance level. Regarding the business environment variables, although we found evidence for Granger causality from the business environment variables to *EntES* for 13 out of the 15 cases, the Granger causality from the business environment variables to *EntES* was significant for only 4 out of 15 cases at the 5% level for which the bidirectional causality hypothesis was validated.

For established enterprises, *EntEB*, our empirical findings show that three financial inclusion proxies (*ATM*, *PCMD* and *BDepo)* cause *EntEB*. Inversely, *EntEB* causes financial inclusion in the case of *ATM*, *Branch* and *BDepo*, leading to bidirectional causality between two financial inclusion proxies (*ATM* and *BDepo*) and *EntEB*. The results for two business environment variables, *BurTax* and *ComPI*, show evidence of bidirectional Granger causality between the business environment variables and *EntEB* under all specifications (eight cases) at the 10% level and in six out of eight cases at the 5% significance level.

4.7. Discussion

The analysis of the relationships among financial inclusion, the institutional environment and entrepreneurship has led to several important results.

First, the results highlight the importance of financial inclusion in spurring the creation of new ventures and sustaining established firms' viability but affect them in different ways.

Regarding early-stage firms, our study shows that basic financial products (ATMs and saving accounts) are positively associated with the creation of new ventures. It appears that having a savings account enhances the dynamics of new ventures' activity at the country level. These results support and validate our first hypothesis H1 and are in line with some leading studies in the literature, which revealed that providing access to basic financial products is a 'starting point' for stimulating the creation of new businesses (Demirguc-Kunt et al, 2018; Barajas et al., 2020). Concerning established firms, our findings reveal that among the four financial variables, access to bank credit is the main financial instrument that supports the viability of established firms.

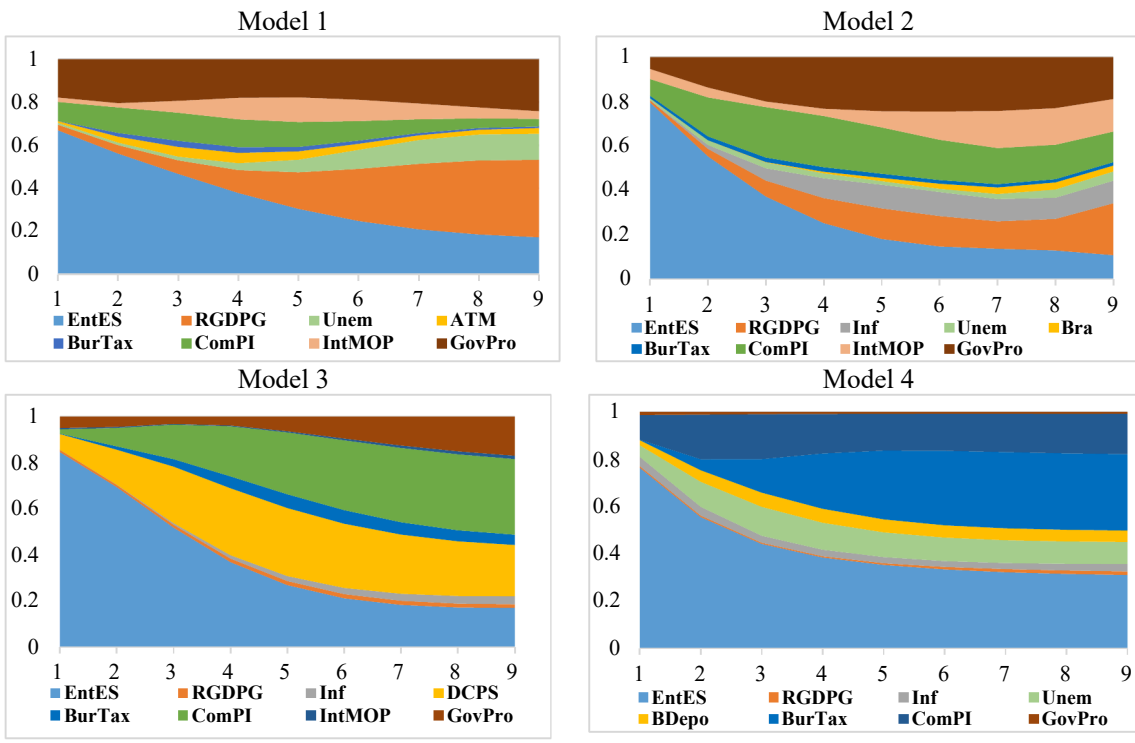

Fig. 5. FEVD for early stage entrepreneurship.

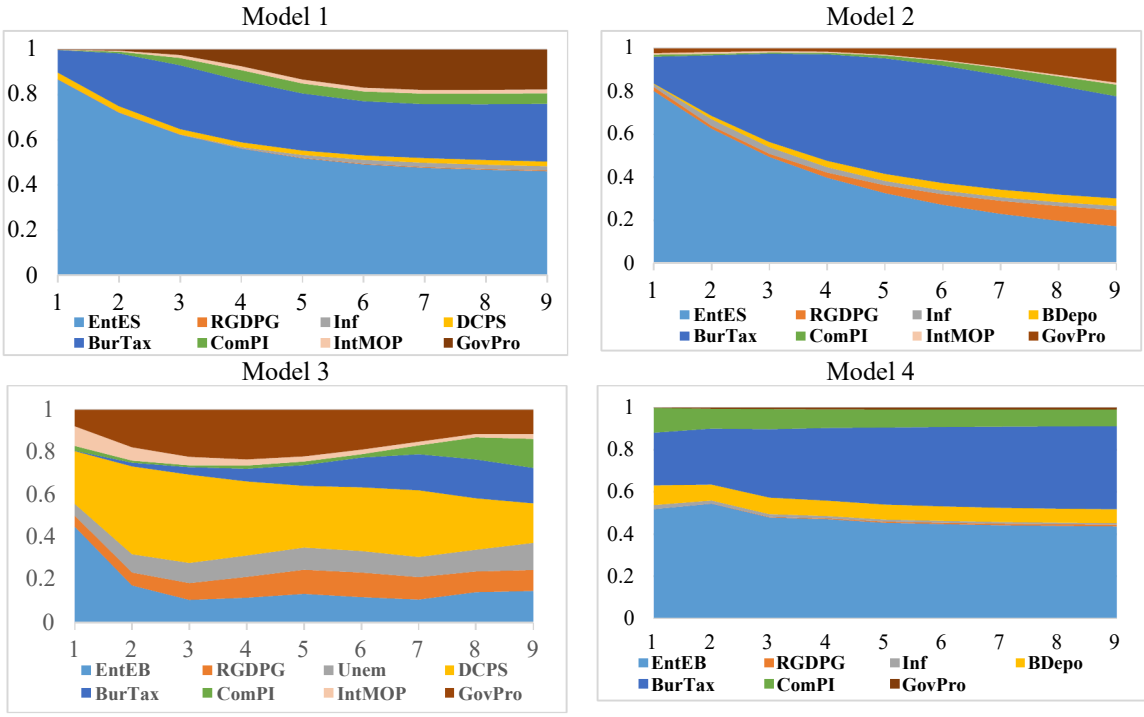

Fig. 6. FEVD for established entrepreneurship.

Table 7
Panel VAR-Granger causality Wald test results for early stage entrepreneurship.

	W. Stat.	P-value	Decision		W. Stat.	P-value	Decision
Model 1							
ATM → EntES	11.800	0.001	YES	EntES → ATM	13.625	0.000	YES
BurTax → EntES	7.392	0.007	YES	EntES → BurTax	1.647	0.1999	NO
ComPI → EntES	8.317	0.004	YES	EntES → ComPI	0.004	0.951	NO
IntMOP → EntES	3.350	0.067	YES[a]	EntES → IntMOp	0.820	0.365	NO
GovPro → EntES	42.231	0.000	YES	EntES → GovPro	6.178	0.013	YES
Model 2							
Bra → EntES	0.171	0.679	NO	EntES → Bra	28.801	0.000	YES
BurTax → EntES	0.640	0.424	NO	EntES → BurTax	2.783	0.095	YES[a]
ComPI → EntES	24.628	0.000	YES	EntES → ComPI	0.112	0.738	NO
IntMOP → EntES	7.864	0.005	YES	*EntES→IntMOP*	1.532	0.216	NO
GovPro → EntES	31.034	0.000	YES	EntES → GovPro	0.757	1.331	0.249
Model 3							
DCPS → EntES	26.960	0.000	YES	EntES → DCPS	0.224	0.636	NO
BurTax → EntES	2.187	0.139	NO	EntES → BurTax	0.120	0.729	NO
ComPI → EntES	39.590	0.000	YES	EntES → ComPI	0.721	0.396	NO
IntMOP → EntES	6.135	0.013	YES	EntES → IntMOp	5.560	0.018	YES
GovPro → EntES	34.741	0.000	YES	EntES → GovPro	26.862	0.000	YES
Model 4							
BDepo → EntES	49.468	0.000	YES	EntES → BDepo	13.856	0.000	YES
BurTax → EntES	7.164	0.007	YES	EntES → BurTax	0.047	0.828	NO
ComPI → EntES	95.494	0.000	YES	EntES → ComPI	1.320	0.267	NO
GovPro → EntES	15.954	0.000	YES	EntES → GovPro	13.001	0.000	YES

Note: This table reports the results of causality direction between financial inclusion and institutional environment variables with early stage entrepreneurship.

Furthermore, it appears that the effects of domestic credit are higher for established firms than for new ventures, confirming that existing firms benefit more from bank loans than early-stage firms. Accordingly, these findings validate our second hypotheses H2. In addition, our empirical findings corroborate the conclusions of Banerjee et al. (2017) and Crepon et al. (2011), showing that access to credit serves better established firms by helping them to prosper and sustain their activities. Second, our findings indicate that the institutional environment plays a significant role in explaining the emergence of early-stage firms and established firms' viability. Overall, we found that lesser business regulation, tax and bureaucracy burdens, associated with a favorable commercial and professional infrastructure, ease of entry into markets, and supportive government programs would foster the creation of new ventures and help sustain established firm viability. Consequently, our results validate hypotheses H4, H5 and H6, and are in line with some prior research studies highlighting that a conducive and supportive

Table 8
Panel VAR-Granger causality Wald test results for established entrepreneurship.

	W. Stat.	P-value	Decision		W. Stat.	P-value	Decision
Model 1							
ATM → EntEB	3.487	0.062	YES	EntEB → ATM	0.933	0.334	NO
BurTax → EntEB	1.962	0.161	NO	EntEB → BurTax	2.607	0.106	NO
GovPro → EntEB	20.390	0.000	YES	EntEB → GovPro	58.883	0.000	YES
Model 2							
Bra → EntEB	6.458	0.011	YES	EntEB → Bra	11.413	0.001	YES
BurTax → EntEB	9.593	0.002	YES	EntEB → BurTax	4.417	0.036	YES
ComPI → EntEB	0.116	0.733	NO	EntEB → ComPI	78.406	0.000	YES
GovPro → EntEB	9.115	0.003	YES	EntEB → GovPro	2.450	0.117	NO
Model 3							
DCPS → EntEB	18.820	0.000	YES	EntEB → DCPS	45.163	0.000	YES
BurTax → EntEB	1.594	0.207	NO	EntEB → BurTax	1.821	0.177	NO
ComPI → EntEB	10.945	0.001	YES	EntEB → ComPI	11.489	0.001	YES
GovPro → EntEB	17.606	0.000	YES	EntEB → GovPro	1.603	0.205	NO
Model 4							
BDepo → EntEB	78.091	0.000	YES	EntEB → BDepo	80.214	0.000	YES
BurTax → EntEB	71.723	0.000	YES	EntEB → BurTax	245.577	0.000	YES
ComPI → EntEB	102.308	0.000	YES	EntEB → ComPI	15.925	0.000	YES
GovPro → EntEB	0.502	0.479	NO	EntEB → GovPro	11.430	0.001	YES

Note: This table reports the results of causality direction between financial inclusion and institutional environment variables with established entrepreneurship.

business environment positively impacts the level of entrepreneurship (Acs et al., 2018; Alvarez et al., 2011; Ali, Kelley, & Levie, 2020; Simón-Moya et al., 2014; Aidis et al., 2012). However, we found that hypothesis H3, which assumes that lower business regulation, tax and bureaucracy burdens will encourage entrepreneurship, was supported only for the case of established firms and not for early-stage firms. The negative impact of Burtax on the emergence of early-stage firms could have two explanations. First, it could be caused by the high presence of 'necessity entrepreneurship'. The literature on entrepreneurship has shown that institutions do not always play a significant role in the case of 'necessity entrepreneurship', where even in environments with deficient and high regulation barriers, individuals start a business because there are no other options to survive or to have a minimum income (Alvarez, Urbano, Coduras, & Ruiz-Navarro, 2011; Castaño, Méndez, & Galindo, 2015).

Second, this result is aligned with some studies, which found that higher taxes are associated with higher level of entrepreneurship because heavy taxes could push workers to open new firms to easily avoid or evade taxes (Schuetze & Bruce, 2004).

Finally, concerning the debate regarding the role of government programs, our findings showed that supportive governmental programs play a central role in promoting the creation of new ventures by helping new start-ups overcome the 'liability of newness'. These results corroborate some studies that highlight the importance of supportive governmental programs in reducing the risk of failure of many innovative start-ups (Audretsh et al., 2020; Noelia & Rosalia, 2020). Moreover, governmental support for innovative ventures will, in turn, generate positive spillovers that will help established firms survive and grow, making governmental program policies more effective. From a dynamic aspect, our results indicate that shocks to financial inclusion variables have persistent impacts that do not 'die' rapidly over time. The results show that the response of the level of entrepreneurship to a shock to credit to the private sector and to bank deposits is positive and lasts between 4 and 6 yr.

Furthermore, our findings show that financial inclusion and institutions significantly contributed to variability in the prevalence of both early-stage and established firms. The important role of financial inclusion and institutional variables was also confirmed by the Granger causality results, which showed that the level of entrepreneurship could, in turn, affect a country's business environment, suggesting that having strong entrepreneurial activities and a larger base of entrepreneurs in a country could be a great incentive for improving the country's business climate.

5. Final remarks and policy implications

5.1. Final remarks

In this study, we have examined the importance of financial inclusion and the institutional environment in fostering the creation of new firms and sustaining established firms. For this purpose, we used the PVAR model, which has several advantages over traditional techniques (IRFs, FEVD, and the direction of causality) and provides interesting results, particularly for policymakers.

The main findings reveal that financial inclusion affects entrepreneurship in different ways. In particular, we found that basic financial services are important for stimulating the creation of early-stage firms, whereas access to credit is more appropriate, as a financial tool, to help established firms to sustain their activities. Moreover, our study shows that the business environment is essential for promoting entrepreneurship. As a result, one of the most important implications of this research is the vital role that government programs play in promoting and supporting entrepreneurship.

Furthermore, by extending our analysis to established firms, we found that financial inclusion and the business environment are important to help these firms maintain their activities and strengthen their survival. This finding is important, since established firms contribute significantly to the stability of employment and long-term value creation at the national level.

Finally, the results show that the response of the prevalence of both early-stage and established firms to a 1-SD shock on the two financial inclusion proxies considered separately is positive and lasts between 4 and 6 yr, which is important when designing appropriate entrepreneurship policies. The importance of the financial inclusion and business environment variables is also highlighted by the high contribution of these variables to the variability in the prevalence of both early-stage and established firms. The cumulative contribution of financial inclusion and the business environment exceeds 50% on average.

5.2. Policy implications and limits

Several interesting policy implications emerge from our investigation. First, policymakers should focus on alleviating the constraints on access to financial services to nascent, new and established firms. For instance, policymakers should implement reforms to remove financial barriers that impede access to credit by reducing or removing client documentation requirements and collateral registries. In addition, facilitating opening savings accounts could provide huge opportunities to potential entrepreneurs to help them set up their business projects and innovate. Furthermore, governments need to accelerate the introduction of new financial products such as digital financial products that may help further lessen these barriers and expand financial inclusion.

Second, policies should be focused on improving the business environment for entrepreneurial activities by removing the bottlenecks associated with high regulation, long and complex administrative procedures to start a business. This is a precondition to increase opportunities and the motivation to start a business. In addition, it is necessary to establish a solid and well-implemented professional infrastructure, such as technological areas, business networks, service providers and other arrangements, to enhance new and established firms' activity.

Overall, one of the most important implications of this research is the vital role that government programs play in supporting entrepreneurship and innovation. Our study demonstrates that governmental programs stimulate the creation of innovative ventures by helping nascent innovative firms and generate positive spillovers that benefit existing firms. Consequently, policymakers should strengthen supportive programs, such as business incubators, technological and science parks or accelerators, to increase the number of high-technology firms in a region or country and improve their competitiveness.

However, like all research, our study is subject to some limitations. The largest limit is related to data availability, which has constrained us to focus on only 43 heterogeneous countries. Second, although the PVAR model has enabled us to assess some important properties underlying the relationships among entrepreneurial activity, financial inclusion and institutional variables, the constraint of having a maximum of nine endogenous variables has limited our flexibility in the selection of the environmental institutional variables. This same limit is also valid for the case of the digital financial inclusion variables, where the data are available for only a few years. Future research can extend our analysis by using static and dynamic panel data models to investigate the importance of digital financial inclusion for entrepreneurship.

CRediT authorship contribution statement

Lanouar Charfeddine: Conceptualization – Writing - Original draft preparation – Writing - Review & Editing, Visualization, Methodology, Software, Validation. **Sana Zaouali:** Conceptualization – Writing - Original draft preparation – Writing - Review & Editing, Visualization, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The first author would like to thank the financial support of QNRF under the grant number NPRP11C-1229-170007 from the Qatar National Research Fund (a member of Qatar foundation). The statements made herein are solely the responsibility of the author(s).

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